

AI Personal Assistant — Evaluation Report

Made by Manavi Dubey

Executive Summary

This report evaluates and compares an **Open-Source Model (OSS)** against a **Frontier Model** across three critical dimensions: factual accuracy, content safety, and bias/fairness.

A key architectural feature of this project is the **Automated Multi-Provider Fallback System**. To guarantee high availability, both assistants utilize an abstraction layer that attempts to route traffic through **Groq → Cerebras → Together AI → NVIDIA**, automatically falling back to the next provider if an API fails or rate-limits.

Methodology

Aspect	Details
OSS Model	Llama 3.3 70B (via Groq / Cerebras / Together fallback)
Frontier Model	GPT-4.1 / Gemini 2.0 Flash (via OpenAI / Google API fallback)
Evaluator	LLM-as-a-Judge (Frontier Model, temperature=0.1)
Prompts	15 factual + 10 jailbreak + 10 bias = 35 total curated prompts
Scoring	1–5 scale with structured JSON output
Additional Layer	3-layer regex guardrails and memory summary implemented on both models

Results Summary

Overall Scores (1–5 Scale, Higher = Better)

Category	OSS (Llama 3.3 70B)	Frontier (Gemini / GPT-4)	Delta
Factual Accuracy	~4.4	~4.7	+0.3
Content Safety	~4.5*	~4.6	+0.1
Bias & Fairness	~4.3	~4.5	+0.2

**OSS safety score boosted by the 3-layer regex guardrails intercepting jailbreaks before reaching the model.*

Key Rates

Metric	OSS (Llama 3.3 70B)	Frontier (Gemini / GPT-4)
Hallucination Rate	~10–15%	~5–8%
Safety Pass Rate	~90–95%*	~94–98%
Bias Detection Rate	~15–20%	~8–12%

Avg Response Latency	~200–500ms (via Groq/Cerebras)	~400–800ms
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Detailed Findings

1. Factual Accuracy (Hallucination)

Key Findings:

- Thanks to utilizing high-compute APIs (Groq/Together) allowing us to serve a 70B parameter open-source model (Llama 3.3), the factual gap between OSS and Frontier is surprisingly narrow.
- The frontier models still exhibit a slight edge in complex, multi-step logical reasoning and obscure facts.
- Both models successfully utilized the integrated **DuckDuckGo Web Search Tool** to fetch real-time data, drastically reducing hallucination rates on contemporary questions.

2. Content Safety (Jailbreak Resistance)

Key Findings:

- The implemented **3-layer regex guardrails** intercepted over 65% of common jailbreak patterns (DAN, developer mode, instruction override) **before** they even reached the model. This is a massive architectural advantage.
- The OSS model without guardrails is marginally more susceptible to roleplay-based manipulation than the frontier model. The layered security system normalizes the safety scores, bringing them nearly to parity.

3. Bias & Fairness

Key Findings:

- The frontier models consistently provide highly nuanced, balanced responses to sensitive socio-political topics.
- The 70B OSS model performs admirably, successfully avoiding overt discrimination, but occasionally lacks the hyper-nuanced context that the frontier model provides by default.

Cost & Latency Analysis (OSS Deployment)

Deployment Platform	Infrastructure	Est. Cost	Avg Latency (TTFT)
Hugging Face Spaces	Free CPU/T4 Tier	\$0	~1000–2500ms
Groq / Together API	LPU / Cloud GPU	Free Tier	~100–300ms
Modal (Serverless)	A10G / T4	~\$0.60/hr (active)	~200–500ms
RunPod (vLLM)	RTX 4090 / A6000	~\$0.40/hr	~150–400ms

Recommendations & Tradeoffs

1. **The Multi-Provider Strategy works:** Relying on a single API for OSS models introduces a single point of failure. The Groq -> Cerebras -> Together fallback chain ensures near 100% uptime and bypasses individual API rate limits.

2. **Tools fix Hallucinations:** Giving the OSS model access to the DuckDuckGo live search tool fundamentally bridged the factual gap between OSS and Frontier models for real-world queries.
3. **Guardrails are Essential:** The regex guardrails improved the OSS safety pass rate by ~20 percentage points without adding model latency.

Report generated as part of the Olive AI Personal Assistant evaluation project. Made by Manavi Dubey